Tester Engineering Suite v1.0.4-2026

Developed by Tester Present Specialist Automotive Solutions

Professional Automotive Diagnostic & ECU Flash Programming Platform

Multi-Protocol | Multi-Manufacturer | J2534 PassThru Compatible

20+	100+	14	40+	18
Manufacturers	ECU Variants	Protocols	Security Algos	UDS Services

The Complete Diagnostic & Flash Programming Solution

Tester Engineering Suite is a professional-grade automotive diagnostic and ECU flash programming platform built for performance tuners, independent workshops, and automotive engineers. Covering 20+ vehicle manufacturers, 100+ ECU variants, and 14 communication protocols, it delivers everything you need in a single, unified application -- from live OBD2 data monitoring and immobiliser key programming to full ECU reflashing with advanced security bypass capabilities and automatic checksum correction.

Built on the industry-standard J2534 PassThru API, Tester Engineering Suite works with any compatible hardware adapter (Tactrix OpenPort, AVDI, Drew Technologies, and more). The platform also includes a remote reprogramming system that tunnels J2534 PassThru commands over WebSocket via a cloud relay server, enabling technicians to perform full ECU diagnostics and flash programming on vehicles at any location with an internet connection.

- Single Platform -- Replace multiple OEM tools
- Multi-Protocol -- CAN, K-Line, UDS, KWP, VPW+
- J2534 Compatible -- Any PassThru adapter
- Professional Flash -- Auto-checksum correction
- Security Bypass -- 40+ seed/key algorithms
- Live Dashboard -- Configurable gauges & logging
- Immobiliser -- PATS key programming
- Remote Reprogramming -- Cloud J2534 tunnel

Core Features

Live Data & Monitoring

- 100+ standard OBD2 PIDs with continuous polling
- 1000+ OEM PIDs across 8 manufacturer databases
- Custom PID definitions with formula editor
- Real-time multi-PID graphing with zoom/pan
- Professional FPV-style analog gauge dashboard
- **Configurable aux gauges (right-click to change PID)**
- Peak hold recording (min/max/avg/std-dev)
- Histogram distribution analysis
- Data logging with auto-save and CSV export
- Variable-speed data playback (0.5x-4x) with seek
- **Dashboard info overlay (Live Diag / Log Playback)**

Diagnostic Trouble Codes

- OBD2 Stored, Pending, and Permanent DTCs
- UDS ReadDTCInformation with full status masks
- KWP2000 legacy DTC support
- Freeze frame snapshot data
- DTC severity categorisation and colour coding
- Ford DTC database (2+ million entries)
- Clear DTCs with confirmation
- **Global multi-ECU DTC scan via network topology**

CAN Bus Analysis

- Real-time CAN sniffer (dump + unique ID modes)
- Frame filtering (include/exclude by CAN ID)
- Custom CAN frame transmission
- Standard (11-bit) and extended (29-bit) support
- Frame statistics and FPS monitoring
- Raw PassThru multi-protocol access

ECU Flash Programming

- Multi-manufacturer flash support (20+ platforms)
- Pre-flash checksum verification and auto-correction
- **Spanish Oak VID + Data checksum auto-correction**
- **Auto-correct checksums before write (toggle)**
- Block-level or full-flash programming modes
- Calibration-only flashing for quick tune updates
- Post-flash verification and dependency checks
- Progress tracking with real-time transfer speed
- Battery voltage monitoring during flash
- RSA signature recalculation (BMW)
- Safe abort with status recovery guidance

Security & Authentication

- SA2 bytecode interpreter (VW/Audi/Skoda/SEAT)
- GGDS seed/key algorithm (Ford)
- PcmHammer integration (GM)
- LFSR key generation (Mazda)
- VKeyGen DLL integration (Daimler/Mercedes)
- PIN brute-force (PSA Group)
- JLR pre-calculated key tables (311 modules)
- Honda pre-calculated keys (1000+ keys)
- UnlockECU Library -- 40+ additional algorithms
- Ford CAN Generic KeyGenMkI (300+ module keys)
- Hyundai SMARTRA PIN calculator
- Volvo LFSR security (24/64-bit)

Emissions & Readiness

- Complete I/M readiness flag monitoring
- Drive cycle completion verification
- Mode 05 & 06 detailed test results
- Mode 08 on-board test execution
- MIL (check engine light) status display
- **Visual readiness monitor LED panel**
- **Emissions report export**

New in v1.0.4-2026

Immobiliser / PATS Key Programming

A complete BA Falcon Passive Anti-Theft System (PATS) parameter reset implementation, added as a new tab within Diagnostic Services. Supports reading programmed key count from the BEM, monitoring PCM security state transitions, and executing the full 7-step parameter reset (encryption key learn) procedure with real-time progress tracking.

- CAN IDs: 0x6F6 (PCM) / 0x6F8 (BEM) -- 8-byte frames, 500ms cycle
- Full PATS crypto algorithm: addition with carry, multiply, rotate, XOR, nibble swap
- DIDs: 0xC104 (key count from BEM), 0xC115 (security state from PCM)
- Services: 0xB1 01 9C (initiate), 0xB1 00 0B 20 (start EEC key learn)
- 10-minute polling cycle with automatic state detection (0x33/0x55/0xAA)
- TesterPresent keepalive loop during parameter reset procedure

FG Falcon EEPROM Editor (Mk1 + Mk2/FGX)

Full-featured EEPROM field editor for the Ford FG Falcon VDO Instrument Panel Cluster, supporting both Mk1 (512 bytes) and Mk2/FGX (1024 bytes) EEPROM formats. All features from James' reference FGmk1EEPROMTool have been integrated with automatic CRC-CCITT checksum recalculation across all EEPROM regions.

- VIN editing (17-character, auto-uppercase, with copy regions)
- Odometer decode/encode (complex subtraction cipher with dual lookup tables)
- Speedo range selection: 220 km/h, 260 km/h, 260 km/h FGX calibration
- Tacho range selection: 7,000 RPM or 8,000 RPM calibration tables
- FPV features: Splash Screen, Extra Needles, Gauge Type (Oil/Boost), Boost Cal (1/1.5/2 Bar)
- FPV Shift Alert with configurable RPM threshold (0-8000, 50 RPM steps)
- Police Mode toggle (Ford Police Interceptor configuration)
- +2 km/h Speedo Override (digital speedometer correction)
- CRC-CCITT checksum validation: 15 regions (Mk1), 22 regions (Mk2/FGX)
- Hex grid editor with colour-coded EEPROM regions

Spanish Oak PCM Checksum Correction

Dual-checksum verification and auto-correction for Ford EEC-VI Spanish Oak PCMs, verified against PCMflash, PTdiag, and actual PCM behaviour. Checksums can be verified on BIN load and automatically corrected before flash write operations.

- VID Block Checksum (0x100C0-0x1013B) -- 16-bit big-endian sum, (-sum)+1 formula
- Data Checksum (0x10222) -- section-table-based, 32-bit int + (int>>16) summation
- 'Auto-correct checksums before write' toggle checkbox
- 'Verify / Correct Checksums' button with detailed mismatch reporting
- Colour-coded status display: green (OK) / red (MISMATCH)

Configurable Dashboard Gauges

The 6 auxiliary gauges on the FPV-style instrument cluster dashboard are now fully configurable via right-click context menu. Any gauge slot can be reassigned to display any available OBD2 PID, with automatic min/max/warning/danger threshold calculation.

- Right-click any side gauge to select from all 68 Mode 01 PIDs
- Dark-themed context menu with current assignment indicator
- Dashboard info overlay when no data source is active
- Data routing from both Live Diagnostics and Log Playback

Additional Enhancements

- **Single-instance protection** -- Global named mutex prevents running two copies
- **Tools menu** -- Full navigation menu (File | Tools | Help) with all 10 tab shortcuts
- **6HP26 flash fixes** -- Correct read address range (0x000000-0x100070), session 0x81, probe reads
- **6HP26 adaptation read** -- Corrected 0xA1/0xA0 rapid data protocol with burst-mode response capture
- **UI layout fixes** -- Connection panel, Remote panel, Emissions banner, Dashboard LCD spacing

Remote Diagnostic & Programming Technology

Tester Engineering Suite's proprietary remote reprogramming system is an industry-first solution that tunnels the complete J2534 PassThru API over the internet via a secure cloud relay. This enables automotive technicians to perform full ECU diagnostics, flash programming, and fault code management on vehicles at any location worldwide -- as if they were physically connected to the vehicle's OBD2 port.

Three-Tier Cloud Architecture

Master Console	Cloud Relay Server	Slave Agent
Remote technician workstation JSON-RPC 2.0 over WSS	Azure WebSocket Relay Session routing + HMAC signing	On-site J2534 device Connected to vehicle OBD2 port

Secure Session Pairing

- Cryptographically random 6-character session codes
- 10-minute expiry, extended to 60 minutes on pairing
- HMAC-SHA256 response signing (anti-MITM)
- 30-second heartbeat keep-alive mechanism

Full J2534 Tunneling

- All 14 protocols supported remotely
- Remote device enumeration and connection
- Raw CAN frame send/receive with filtering
- ELM327 shim layer for non-native hardware

Remote Flash Programming

- Chunked firmware transfer (65KB) with SHA256 verification
- 12-step UDS flash sequence with vendor extensions
- Real-time progress notifications
- Safe abort with 600-second timeout protection

No VPN. No port forwarding. No direct IP connection required. The cloud relay server handles all session routing -- both master and slave connect outbound, making the system firewall-friendly and deployable anywhere with an internet connection.

Supported Manufacturers & ECU Types

Manufacturer	ECU Types	Capabilities
Volkswagen Group (VW/Audi/Skoda/SEAT)	Simos 8/10/12/12.2/16/18/18.10/18.41 DQ250 MQB, DQ381, Haldex	FRF, BIN, ODX Full unlock + re-lock
Ford	EEC-VI (Spanish Oak) 16 modules Focus, Fiesta, Mondeo, Falcon, Transit, Ranger, Mustang, F-150	VBF format, SBL bootloader 300+ module keys VID + Data checksum correction
Ford (IPC)	FG Mk1/Mk2/FGX VDO Cluster BA Falcon PATS BEM/PCM	EEPROM R/W, FPV features Immobiliser key programming Police Mode, Speedo/Tacho cal
BMW	MS41/42/43/45, MSS70, MSV70 BMS46 (ABS)	BIN format, dual-memory RSA signature recalculation
Honda	Multiple ECU types	RWD + BIN, CAN-based 1000+ pre-calculated keys
Subaru	Denso (CAN + K-Line) Hitachi, M32R	BIN w/ encryption SSM2 diagnostics, 300+ keys
General Motors	PCM (Powertrain) IPC (Instrument Panel)	VPW protocol PcmHammer integration
Daimler (Mercedes)	Multi-block ECUs C/E/S-Class, GLA, GLC	VKeyGen DLL security 100+ security DLLs
ZF Transmissions	6HP26 (6-speed auto TCM)	960KB flash + 128KB cal Adaptation read/reset
Volvo	P1/P2/P3/SPA, P80	SBL bootloader, VBF + Intel HEX D2 CAN protocol, LFSR security
Hyundai / Kia	SIM2K (Infineon)	Full flash, EEPROM SMARTRA PIN calculator
Mazda	RX-8 Denso SH7055	BIN format LFSR key generation
PSA (Stellantis)	VD46/56, CMM, SIM82	PIN-based security Brute-force support
JLR (Jaguar/Land Rover)	311 ECU module groups	368 security constants Multi-level security
Saab	Trionic T5.2/5.5/T7/T8	BIN + checksum correction XOR ByteCoder encryption
Aston Martin	V8/V12 (Ford EEC-VI)	VBF format Shared Ford platform
Bosch (Generic)	ME7.1/7.5/7.9+	K-Line programming McMESS 9-bit calibration

Communication Protocols

Protocol	Description & Capabilities
CAN Bus (ISO 15765)	11-bit & 29-bit addressing, multi-frame transport, raw sniffer, configurable TX/RX IDs
UDS (ISO 14229-1)	Full service support: Session Control, Security Access, Read/Write DID, Memory R/W, Flash, Routine Control, DTC Management, IO Control
KWP2000 (ISO 14230)	Standard/Programming/Extended sessions, seed/key auth, K-Line diagnostics
OBD2 (ISO 15031)	Modes 01-0A: Live data, freeze frames, DTCs, readiness, vehicle info, permanent DTCs
OBD-on-UDS	Seamless OBD2 data via UDS services (0x22 ReadDID, 0x19 ReadDTC)
K-Line (ISO 9141-2)	Physical layer for KWP2000/KWP1281, 5-baud init, fast init
KWP1281	VW legacy K-Line protocol for instrument cluster EEPROM programming
Ford-9141	Ford proprietary K-Line (10,400 baud), no init required, 8-byte frames
VPW (J1850)	GM Variable Pulse-Width K-Line protocol, PcmHammer integration
J1850 PWM	Ford Pulse-Width Modulated protocol, 41.6 kbps
SSM2 (Subaru)	Subaru Select Monitor 2: parameter/switch/DTC reading, K-Line & CAN
McMESS (Bosch)	9-bit K-Line calibration protocol, 10.4k/125k baud, RAM access
Volvo D2	Volvo proprietary CAN diagnostic protocol for P1/P2/P3/SPA platforms
J2534 PassThru API	Multi-protocol hardware abstraction: filter management, raw I/O, voltage control

Diagnostic Services (18 UDS Sub-Tabs)

Tab	Service	Description
Diagnostic Routine Builder	TDR	Drag-and-drop UDS routine sequencing
Session Control	0x10	9 session types + custom, P2/P2* timing
ECU Reset	0x11	5 reset types with timeout control
Tester Present	0x3E	Keep-alive with auto-send interval
Security Access	0x27	22+ algorithms, auto unlock workflow
Communication Control	0x28	Enable/disable message filtering
Read DID + Bruteforcer	0x22	DID reading + automated discovery
Write DID	0x2E	Multi-format write with verify
DTC Management	0x19/0x14	Read/clear/snapshots + extended data
Memory Access	0x23/0x3D	Block read/write, configurable format
IO Control	0x2F	Actuator control by DID
Routine Control	0x31	Module-specific routines + results
ECU Flash	0x34-0x37	Multi-platform flash programming
Raw UDS	Hex	Freeform hex request/response
KWP2000 Legacy	ISO 14230	Legacy K-Line diagnostic support
Bosch McMESS	9-bit K-Line	ME7 calibration, RAM access
Ford ISO9141	Proprietary	10,400 baud, no-init K-Line
Immobiliser / PATS	BA PATS	Key programming + parameter reset

Supported File Formats

Format	Type	Manufacturers	Read	Write
BIN	Raw binary ECU dump	Honda, Nissan, Subaru, GM, BMW, ZF, ME7, etc.	Yes	Yes
FRF	VW encrypted firmware	VW/Continental (Simos series)	Yes	Yes
VBF	Ford/Volvo firmware blocks	Ford, Volvo, Aston Martin	Yes	Yes
RWD	Honda-specific firmware	Honda	Yes	Yes
ODX	Open Diagnostic Data Exchange	VW (export)	Yes	Yes
CAL	Calibration data (128KB)	ZF 6HP26	Yes	Yes
Intel HEX	Bootloader format	Volvo P1 SBL	Yes	--
CSV	Data export	All (logging/export)	--	Yes
EEPROM	Instrument cluster data	Ford FG Mk1 (512B) / Mk2 (1024B)	Yes	Yes

Advanced Features

VW/Simos ECU Unlock & FRF Tools

- CBOOT patching (RSA bypass via NOPs)
- Write-without-erase exploit (ECC-correct)
- FRF decryption (recursive XOR cipher)
- FRF export to ODX or BIN
- Custom AES-128-CBC key/IV override
- DSG cipher (DQ250 MQB gearbox)
- Multi-mode: Standard, CAL Only, Unlock, Full, Re-Lock

DID Bruteforcer

- Automated data identifier discovery
- Configurable range and delay
- Live NRC logging and analysis

ZF 6HP26 Transmission

- 960KB full flash read/write
- 128KB calibration read/write
- Adaptation data read (0xA1/0xA0 rapid data)
- Adaptation reset with backup/restore
- Fuzzy CalID matching
- Session 0x81 + probe reads before flash

Instrument Cluster Programming

- FG Mk1/Mk2/FGX EEPROM editor (VIN, odo, FPV)
- FG2 IPC full flash (256KB NEC V850)
- Custom splash image upload (13 presets + BMP)
- KWP1281 cluster programming (VW legacy)
- GM IPC flash programming

Application Structure (12 Main Tabs)

Tab	Description
ECU Identification	VIN decoder, Mode 09 vehicle info, UDS ECU identification, supported PID query
Network Topology	Automated multi-ECU discovery across CAN/K-Line buses, global DTC scan
DTC Memory	OBD2 stored/pending/permanent DTCs, UDS ReadDTCInformation, freeze frames
Emissions Tests	I/M readiness monitors, Mode 05/06/08, visual LED panel, report export
Dashboard	FPV-style cluster: RPM, Speed, Coolant, Fuel + 6 configurable aux gauges
Live Diagnostics	Standard + Custom + OEM PID polling, real-time data grid, configurable rates
Signal Analyser	Multi-PID graphing with zoom/pan, histogram distribution, statistical analysis
Diagnostic Services	18 UDS service sub-tabs (see protocol table), ECU flash, immobiliser
Data Player	Log replay with play/pause/seek, variable speed, distribution analysis
CAN Sniffer	Protocol-agnostic traffic capture, filtering, statistics, frame transmission
PassThru Control	Raw J2534 API: connection, messaging, config, init, filters, voltages
License Management	License key, machine ID, server verification, key export

System Requirements

Component	Minimum	Recommended
Operating System	Windows 7	Windows 10/11
Runtime	.NET Framework 4.8.1	.NET Framework 4.8.1
RAM	512 MB	2 GB+
Disk Space	50 MB + log storage	200 MB+ (with logs)
Hardware Interface	Any J2534 PassThru device	Tactrix OpenPort 2.0 / AVDI
Connection	USB (J2534 adapter)	USB 2.0+ / Internet (remote)
Display	1100 x 750	1920 x 1080+ (for dashboard)

Why Choose Tester Engineering Suite?

One Tool, Every Vehicle

Stop juggling multiple OEM diagnostic tools. 20+ manufacturers, 100+ ECU variants, and 40+ security algorithms in a single application.

Professional Flash Programming

Full ECU read, write, and unlock with automatic checksum correction, block-level verification, and comprehensive safety warnings.

Hardware Freedom

Built on J2534 PassThru standard. No proprietary hardware lock-in -- choose the interface that fits your budget and workflow.

Built by Engineers, for Engineers

Developed by automotive engineers with real workshop experience. Every feature designed for practical diagnostic and tuning workflows.

Ready to Transform Your Workshop?

Visit **testerpresent.com.au** for licensing, documentation, and support

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